

KETHNI NANAYAKKARA

Software Engineering Undergraduate | Open Source Contributor | AI/ML Enthusiast

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PROFESSIONAL SUMMARY

Software Engineering undergraduate at NSBM Green University (University of Plymouth, UK) with merged contributions to WSO2 production repositories, co-authorship of a published conference research abstract, and a real-time AI-powered Chrome extension available on the Chrome Web Store. Experienced in full-stack, mobile, and AI/ML development through academic, research, and open-source projects.

EDUCATION

BSc (Hons) Software Engineering | NSBM Green University - University of Plymouth, UK | 2023 – 2026

Foundation Programme in Computing | NSBM Green University | 2023 – Oct 2023

PROJECTS

EmoUI - Emotion-Adaptive Chrome Extension | 2025 – 2026

Individual Final Year Project | [Live on Chrome Web Store](#)

Python · React · Tailwind CSS · OpenCV · YOLO11n-cls · MediaPipe

Built a real-time, privacy-first, on-device facial emotion recognition Chrome extension with adaptive UI themes, audio feedback, and motivational quotes in English and Sinhala. Trained and benchmarked three models (YOLO11n-cls, custom CNN, MediaPipe) on a multi-source dataset, achieving 94.2% Top-1 accuracy with YOLO11n-cls. Published to the Chrome Web Store; associated conference paper submitted to ICTAR.

Hotel Booking & Reservation System | 2026

Full-Stack Group Project

React · Node.js · Express.js · MongoDB

Built a full-stack platform for hotel listings, room availability, and reservation management using collaborative Git workflows and Agile practices.

FurniViz - Interactive Furniture & Room Visualizer | 2026

Group Project | Role: UX Lead

React.js · Vite · Three.js · Fabric.js · Tailwind CSS

Built an interactive web app with a real-time synchronized 2D grid planner and 3D viewer for room layout design. Designed interaction flows, applied Nielsen's usability heuristics, and conducted think-aloud usability testing.

Caring for Carers - Autism Caregiver App | 2025

Mobile Group Project + Research

Flutter · Firebase · TensorFlow Lite · Gemini API

Developed stateful UI components and a visual communication library with Firebase real-time integration. Contributed to an AI-assisted caregiver support system and co-authored a published conference abstract at KDU IRC 2025.

GlowMate - Skincare Recommendation App | 2025

Flutter · Firebase Authentication · Firestore

Implemented user authentication and Firestore-backed quiz result handling.

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js

Databases: MongoDB, MySQL, Firestore

AI / ML: CNN, YOLO (Ultralytics), MediaPipe, TensorFlow Lite, OpenCV, scikit-learn

Tools & Practices: Git, GitHub, Agile/Scrum

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor | [WSO2 \(GitHub: KethniiImasha\)](#) | 2025 – Present

identity-apps #10345 (Merged) - Improved API resource dropdown UX; added noApiResourcesAvailable i18n key and refactored nested ternary logic across 7 files.

docs-is #6174 (Merged) - Streamlined Asgardeo onboarding flow; removed redundant steps and updated regional Console URLs.

docs-is #5916 (Merged) - Fixed grammar and standardized technical casing in SMS/Email OTP docs and the Identity Server Quick Start Guide.

docs-is #5828 (Merged) - Corrected spelling errors and broken curl URL path placeholders in the Asgardeo Identity Governance API spec.

docs-apim #11470 (Merged) - Fixed MkDocs list rendering in the API Manager Quick Start Guide.

identity-apps #10406 (In Review) - Replaced non-semantic `<span role="button">` with a native `<button>` element in ActiveFiltersBar for improved accessibility and SCSS reset applied to preserve styling.

identity-apps #10370 (In Review) - Fixed react-hooks/rules-of-hooks ESLint violation in form-fields-helper.tsx by replacing misused useState with a plain variable.

RESEARCH & PUBLICATIONS

ICTAR (Submitted Conference Paper): Culturally Diverse, Demographically Aware Facial Emotion Recognition Using YOLO11n-Classification: A Comparative Study - Associated with the EmoUI final-year project.

KDU IRC 2025 (Published Conference Abstract): AI-assisted caregiver support system — co-author; contributed to system design and evaluation.

REFERENCES

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